

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{1}{4} m_1 \tilde{\mu} g_1^4 y_d^{i1i2} \text{LF}_{2,2,0}[\mathbf{m}_1, \tilde{\mu}] + \frac{1}{12} m_2 \tilde{\mu} g_2^4 (9 y_d^{i1i2} - 4 y_d^{pi2} \delta_{pi1}) \text{LF}_{2,2,0}[\mathbf{m}_2, \tilde{\mu}] + \right. \\ & \frac{1}{6} m_2 \tilde{\mu} g_2^4 y_d^{pi2} \text{LF}_{3,2,-1}[\mathbf{m}_2, \tilde{\mu}] \delta_{pi1} + \frac{1}{18} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} (3 y_d^{i1i2} - y_d^{si2} \delta_{si1} - 2 y_d^{i1s} \delta_{si2}) \\ & \text{LF}_{2,2,0}[\mathbf{m}_d^r, \mathbf{m}_q^p] + \frac{1}{36} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2}) \text{LF}_{3,2,-1}[\mathbf{m}_d^r, \mathbf{m}_q^p] + \\ & \frac{1}{18} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} (3 y_d^{i1i2} - y_d^{si2} \delta_{si1} - 2 y_d^{i1s} \delta_{si2}) \text{LF}_{2,2,0}[\mathbf{m}_e^r, \mathbf{m}_l^p] + \\ & \frac{1}{36} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2}) \text{LF}_{3,2,-1}[\mathbf{m}_e^r, \mathbf{m}_l^p] + \\ & \frac{1}{18} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2}) \text{LF}_{3,1,0}[\mathbf{m}_l^p, \mathbf{m}_e^r] + \\ & \frac{1}{18} \tilde{\mu} g_1^2 \overline{y_e^{pr}} a_e^{pr} (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2}) \text{LF}_{3,-1}[\mathbf{m}_l^p, \mathbf{m}_e^r] - \frac{1}{8} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} \\ & (g_2^2 (y_d^{i1i2} + y_d^{si2} \delta_{si1}) + g_1^2 (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2})) \text{LF}_{4,1,-1}[\mathbf{m}_l^p, \mathbf{m}_e^r] + \\ & \frac{1}{18} \tilde{\mu} \overline{y_e^{pr}} a_e^{pr} (3 g_2^2 y_d^{si2} \delta_{si1} + g_1^2 (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2})) \text{LF}_{5,1,-2}[\mathbf{m}_l^p, \mathbf{m}_e^r] + \\ & \frac{1}{18} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2}) \text{LF}_{3,1,0}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \\ & \frac{1}{18} \tilde{\mu} g_1^2 \overline{y_d^{pr}} a_d^{pr} (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2}) \text{LF}_{3,2,-1}[\mathbf{m}_q^p, \mathbf{m}_d^r] - \frac{1}{24} \tilde{\mu} \overline{y_d^{pr}} a_d^{pr} \\ & (9 g_2^2 (y_d^{i1i2} + y_d^{si2} \delta_{si1}) + 5 g_1^2 (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2})) \text{LF}_{4,1,-1}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \\ & \frac{1}{6} \tilde{\mu} \overline{y_d^{pr}} a_d^{pr} (3 g_2^2 y_d^{si2} \delta_{si1} + g_1^2 (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2})) \text{LF}_{5,1,-2}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \\ & \frac{1}{36} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (-9 g_2^2 y_d^{si2} \delta_{si1} + g_1^2 (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2})) \text{LF}_{2,2,0}[\mathbf{m}_q^p, \mathbf{m}_u^r] + \\ & \frac{3}{4} \tilde{\mu} g_2^2 y_d^{i1i2} \overline{y_u^{pr}} a_u^{pr} \text{LF}_{3,1,0}[\mathbf{m}_q^p, \mathbf{m}_u^r] + \\ & \frac{1}{72} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} ((g_1^2 - 9 g_2^2) (3 y_d^{i1i2} - y_d^{si2} \delta_{si1}) - 2 g_1^2 y_d^{i1s} \delta_{si2}) \text{LF}_{3,2,-1}[\mathbf{m}_q^p, \mathbf{m}_u^r] - \\ & \frac{3}{4} \tilde{\mu} g_2^2 y_d^{i1i2} \overline{y_u^{pr}} a_u^{pr} \text{LF}_{4,1,-1}[\mathbf{m}_q^p, \mathbf{m}_u^r] + \\ & \frac{1}{36} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (9 g_2^2 y_d^{si2} \delta_{si1} + g_1^2 (3 y_d^{i1i2} - y_d^{si2} \delta_{si1} - 2 y_d^{i1s} \delta_{si2})) \text{LF}_{3,1,0}[\mathbf{m}_u^r, \mathbf{m}_q^p] + \\ & \frac{1}{36} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (9 g_2^2 y_d^{si2} \delta_{si1} + g_1^2 (3 y_d^{i1i2} - y_d^{si2} \delta_{si1} - 2 y_d^{i1s} \delta_{si2})) \text{LF}_{3,2,-1}[\mathbf{m}_u^r, \mathbf{m}_q^p] - \\ & \frac{1}{12} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (9 g_2^2 y_d^{si2} \delta_{si1} + g_1^2 (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2})) \text{LF}_{4,1,-1}[\mathbf{m}_u^r, \mathbf{m}_q^p] + \\ & \frac{1}{6} \tilde{\mu} \overline{y_u^{pr}} a_u^{pr} (3 g_2^2 y_d^{si2} \delta_{si1} + g_1^2 (-3 y_d^{i1i2} + y_d^{si2} \delta_{si1} + 2 y_d^{i1s} \delta_{si2})) \text{LF}_{5,1,-2}[\mathbf{m}_u^r, \mathbf{m}_q^p] + \\ & \frac{5}{8} m_1 \tilde{\mu} g_1^4 y_d^{i1i2} \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_1] - \\ & \frac{1}{24} m_1 \tilde{\mu} g_1^2 (3 g_2^2 y_d^{pi2} \delta_{pi1} + g_1^2 (-3 y_d^{i1i2} + y_d^{pi2} \delta_{pi1} + 2 y_d^{i1p} \delta_{pi2})) \text{LF}_{4,1,-1}[\tilde{\mu}, \mathbf{m}_1] - \\ & \frac{1}{2} m_1 \tilde{\mu} g_1^4 y_d^{i1i2} \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_1] + \\ & \frac{1}{18} m_1 \tilde{\mu} g_1^2 (3 g_2^2 y_d^{pi2} \delta_{pi1} + g_1^2 (-3 y_d^{i1i2} + y_d^{pi2} \delta_{pi1} + 2 y_d^{i1p} \delta_{pi2})) \text{LF}_{5,1,-2}[\tilde{\mu}, \mathbf{m}_1] + \\ & \frac{1}{3} m_2 \tilde{\mu} g_2^4 y_d^{pi2} \text{LF}_{3,1,0}[\tilde{\mu}, \mathbf{m}_2] \delta_{pi1} + \frac{1}{24} m_2 \tilde{\mu} g_2^4 (51 y_d^{i1i2} + 8 y_d^{pi2} \delta_{pi1}) \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_2] - \\ & \frac{1}{8} m_2 \tilde{\mu} g_2^2 (7 g_2^2 y_d^{pi2} \delta_{pi1} + g_1^2 (-3 y_d^{i1i2} + y_d^{pi2} \delta_{pi1} + 2 y_d^{i1p} \delta_{pi2})) \text{LF}_{4,1,-1}[\tilde{\mu}, \mathbf{m}_2] - \\ & 2 m_2 \tilde{\mu} g_2^4 y_d^{i1i2} \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_2] + \\ & \frac{1}{6} m_2 \tilde{\mu} g_2^2 (3 g_2^2 y_d^{pi2} \delta_{pi1} + g_1^2 (-3 y_d^{i1i2} + y_d^{pi2} \delta_{pi1} + 2 y_d^{i1p} \delta_{pi2})) \text{LF}_{5,1,-2}[\tilde{\mu}, \mathbf{m}_2] + \\ & \frac{1}{18} m_1 \tilde{\mu} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} \text{LF}_{2,1,1,0}[\mathbf{m}_1, \mathbf{m}_d^r, \mathbf{m}_q^p] - \\ & \frac{1}{18} m_1 \tilde{\mu} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_d^r, \mathbf{m}_q^p] + \frac{1}{6} m_1 \tilde{\mu} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} \\ & \text{LF}_{2,1,1,0}[\mathbf{m}_1, \mathbf{m}_d^r, \tilde{\mu}] - \frac{1}{6} m_1 \tilde{\mu} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_d^r, \tilde{\mu}] - \\ & \frac{1}{9} m_1 \tilde{\mu} g_1^4 y_d^{i1i2} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_d^{i2}, \tilde{\mu}] + \frac{1}{12} m_1 \tilde{\mu} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} \text{LF}_{2,1,1,0}[\mathbf{m}_1, \mathbf{m}_q^p, \tilde{\mu}] - \\ & \frac{1}{12} m_1 \tilde{\mu} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_q^p, \tilde{\mu}] - \\ & \frac{1}{36} m_1 \tilde{\mu} g_1^4 y_d^{i1i2} \text{LF}_{3,1,1,-1}[\mathbf{m}_1, \mathbf{m}_q^{i1}, \tilde{\mu}] - \frac{7}{24} m_1 \tilde{\mu} g_1^2 \overline{y_d^{pr}} y_d^{pi2} y_d^{i1r} \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \tilde{\mu}, \mathbf{m}_d^r] - \\ & \frac{7}{18} m_1 \tilde{\mu} g_1^4 y_d^{i1i2} \text{LF}_{2,2,1,-1}[\mathbf{m}_1, \tilde{\$$